



Discontinuation of β -blockers in stable patients with previous myocardial infarction, preserved left ventricular ejection fraction, and no heart failure: a pooled analysis of individual patient data

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Summary

Background Many stable patients with chronic coronary syndrome continue β -blocker therapy for years after a myocardial infarction, despite having no clear ongoing indication. Whether β -blockers can be safely discontinued in such patients remains uncertain. The objective of this analysis was to assess the non-inferiority of β -blocker discontinuation versus continuation in patients with previous myocardial infarction and no remaining indication.

Methods We performed a pooled analysis of individual patient data from the ABYSS and SMART-DECISION randomised trials in stable patients over 6 months after a myocardial infarction with a left ventricular ejection fraction of $\geq 40\%$ and no heart failure. Non-inferiority was assessed for the composite primary endpoint of all-cause death, myocardial infarction, stroke, or hospitalisation for cardiovascular reasons, and for the key secondary endpoint of all-cause death, myocardial infarction, or hospitalisation for heart failure. Analyses used a one-stage mixed-effects Cox proportional hazards model including trial as a random effect with margins of 1.25 for the primary endpoint and 1.40 for the key secondary endpoint. The review protocol was registered with PROSPERO, number CRD420261299839.

Findings Among 6238 patients (3698 from ABYSS and 2540 from SMART-DECISION), 3092 were assigned to β -blocker discontinuation and 3146 to continuation. Median time from index myocardial infarction to randomisation was 3.6 years (IQR 1.6–7.5) and median follow-up was 3.0 years (2.3–3.8). The primary endpoint occurred in 532 (17.2%) of 3092 patients in the β -blocker discontinuation group and 500 (15.9%) of 3146 in the continuation group (hazard ratio 1.09 [95% CI 0.97–1.24]; $p_{\text{non-inferiority}}=0.016$). The key secondary endpoint occurred in 201 (6.5%) of 3092 patients in the discontinuation group and 201 (6.4%) of 3146 patients in the continuation group (1.01 [0.83–1.23]; $p_{\text{non-inferiority}}=0.0006$). Findings were consistent irrespective of background left ventricular ejection fraction.

Interpretation In stable patients with previous myocardial infarction, left ventricular ejection fraction $\geq 40\%$, no heart failure, and no other indication for β -blocker therapy, discontinuation met the non-inferiority criteria for the primary and key secondary endpoints. Interpretation of the primary endpoint is limited by its hospitalisation-heavy and trial-dependent composition.

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Introduction

Worldwide, millions of patients with chronic coronary syndrome are likely to remain on long-term β -blocker therapy for years after a myocardial infarction, often despite having no heart failure, reduced left ventricular ejection fraction (LVEF), or another current indication for treatment. The evidence base for this practice predates the reperfusion era,^{1–3} and has been revised following the findings of several randomised clinical trials conducted since.^{4–8} Four trials focused on early initiation of β -blocker therapy during the initial myocardial infarction management,^{4–6} and two others

addressed the distinct clinical question of discontinuation versus continuation of β -blocker therapy in patients with chronic coronary syndrome and a previous myocardial infarction.^{7,8}

The trials on early initiation of β -blocker therapy have produced heterogeneous results, reflecting limited statistical power at the individual trial level. Two subsequent individual participant data meta-analyses reconciled the results of these recent trials by identifying LVEF as a key modifier of treatment effect.^{9,10} Across these analyses, β -blocker therapy was associated with clinical benefit in patients with mildly reduced LVEF (40–49%), whereas no

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See Online for appendix

Research in context

Evidence before this study

The role of β -blockers after myocardial infarction in the modern reperfusion era has been challenged, particularly in patients with uncomplicated presentations, preserved left ventricular ejection fraction (LVEF), and complete revascularisation. Randomised controlled trials evaluating long-term β -blocker treatment initiated in the early post-myocardial infarction period (<14 days) have reported heterogeneous results. Individual participant data meta-analyses have identified LVEF as a key modifier of treatment effect with benefits extending to patients with mildly reduced LVEF (40–49%) and no evidence of superiority in those with preserved LVEF. However, the distinct clinical question of long-term discontinuation versus continuation in clinically stable patients beyond the acute period has not been addressed. We conducted a systematic review of randomised controlled trials published or completed between Jan 1, 2010, and Jan 19, 2026, evaluating long-term oral β -blocker discontinuation versus continuation in patients with myocardial infarction at least 6 months before and LVEF $\geq 40\%$, by searching PubMed with the terms “beta blockers in myocardial infarction” OR (“Adrenergic beta-Antagonists”[Mesh] AND “Myocardial Infarction”[Mesh]) AND “Clinical trial and Randomized Controlled Trial”, and ClinicalTrials.gov with the terms “Myocardial Infarction” AND “Beta-Blockers” AND “Phase 3,4” AND “Interventional studies”. We pooled individual patient data of the only two randomised trials which met these criteria, ABYSS and SMART-DECISION, to assess the non-inferiority of long-term β -blocker discontinuation in stable patients after myocardial infarction.

benefit was shown in the lower-risk population with preserved LVEF ($\geq 50\%$). However, these analyses addressed early initiation of β -blocker therapy and did not examine the distinct question of long-term discontinuation versus continuation in stable patients. Consequently, the available evidence has not been fully reconciled with the ABYSS trial,⁷ which did not show non-inferiority of β -blocker discontinuation for a composite cardiovascular endpoint, largely due to a higher rate of cardiovascular hospitalisations in the discontinuation group. These findings raised concerns regarding the safety of β -blocker discontinuation in patients with chronic post-myocardial infarction, particularly with respect to hospitalisation-related outcomes and increase in blood pressure.¹¹

The SMART-DECISION trial⁸ addressed a closely related clinical question in a similar stable post-myocardial infarction population, with a composite primary endpoint of death, myocardial infarction, or hospitalisation for heart failure. SMART-DECISION used a randomised, open-label design with blinded endpoint adjudication comparable to that of ABYSS and showed non-inferiority of β -blocker discontinuation, supporting the safety of this strategy in the chronic post-myocardial infarction setting.

Added value of this study

This collaborative pooled analysis addresses the unresolved question of late β -blocker discontinuation in the long-term management of patients with history of myocardial infarction and LVEF $\geq 40\%$. Using individual patient data from the only eligible randomised trials, the patient-level approach allowed consistent time-to-event analyses, assessment of treatment-effect modification, and sensitivity analyses for missing data. By showing non-inferiority for the composite endpoint of death, myocardial infarction, stroke, or hospitalisation for cardiovascular reason, this study provides evidence to inform consideration of a late discontinuation strategy for patients for whom continuation of β -blockers is not otherwise indicated. Although the primary endpoint results might have been driven by cardiovascular hospitalisation, findings were consistent for the key secondary endpoint—a composite of death, myocardial infarction, or hospitalisation for heart failure—which comprised more objectively defined events and was therefore less vulnerable to bias arising from the open-label design.

Implications of all the available evidence

β -blocker discontinuation was found to be non-inferior to a strategy of β -blocker continuation in stable patients after myocardial infarction and LVEF $\geq 40\%$. Taken together, the available evidence supports a stage-specific and individualised approach to β -blocker therapy after myocardial infarction. Late discontinuation after initial treatment in the early post-acute phase can be considered with appropriate clinical monitoring.

After conducting a systematic review of randomised trials evaluating β -blocker discontinuation after myocardial infarction, only two randomised controlled trials, ABYSS and SMART-DECISION, specifically addressed this question. Pooling individual patient data from ABYSS and SMART-DECISION offers a unique opportunity to increase statistical power and reassess the non-inferiority hypothesis shared by both trials.

Methods

Population and data sources

We conducted a pooled analysis of individual patient data from the ABYSS⁷ (NCT03498066) and SMART-DECISION⁸ (NCT04769362) trials. This analysis was registered with PROSPERO (CRD420261299839). The design, procedures, and statistical analysis plans of both trials have been previously reported. The statistical analysis plan of this pooled individual patient data analysis was finalised on Feb 5, 2026, after the SMART-DECISION database was locked and before analysis of the pooled outcome dataset. Before finalisation of the statistical analysis plan, unmasked SMART-DECISION trial-level results and patient-level outcome data were accessible to authors contributing to the pooled analysis plan. The pooled

endpoints and non-inferiority margins were inherited from the parent trials and were not modified after data access.

Both trials assessed the safety of long-term β -blocker discontinuation versus continuation in patients who were stable after a myocardial infarction and had LVEF $\geq 40\%$, using a randomised, open-label, non-inferiority design with blinded adjudication of clinical events. A summary of key inclusion and exclusion criteria of both trials is available in the appendix (p 20). Briefly, in ABYSS, patients currently treated with β -blockers were eligible for enrolment in the absence of any acute coronary syndrome in the previous 6 months, ventricular or supraventricular arrhythmia in the past year, or heart failure in the past 2 years. In SMART-DECISION, patients were eligible at least 1 year after myocardial infarction, provided they had no heart failure or atrial fibrillation. Further inclusion and exclusion criteria are described in the original trial reports.^{7,8}

Both trials were conducted according to appropriate local ethical and institutional requirements and all patients provided written informed consent. Detailed parent-trial ethics approval and oversight information is provided in the appendix (p 25).

Outcomes

The primary endpoint for this analysis was a composite of death from any cause, myocardial infarction, stroke, or hospitalisation for cardiovascular reasons. Hospitalisation for cardiovascular reasons was defined according to each trial's own clinical events committee charter and pooled without cross-trial cause-specific mapping or harmonisation. The key secondary endpoint was the composite of death from any cause, myocardial infarction, or hospitalisation for heart failure. Additional secondary endpoints were the composites of: death from any cause, myocardial infarction, and stroke; death from any cause, myocardial infarction, stroke, or hospitalisation for heart failure; death from any cause or hospitalisation for heart failure; and death from any cause, myocardial infarction, stroke, or coronary revascularisation; as well as the individual components of these composite endpoints. Quality of life was assessed with the use of the EQ-5D-5L questionnaire in ABYSS and the PROMIS-29 in SMART-DECISION.¹² Changes from baseline were evaluated longitudinally. In both trials, all clinical events were adjudicated by independent clinical event committees whose members were unaware of treatment assignments.

Statistical analysis

All analyses were performed according to the statistical analysis plan provided in the appendix (p 30). The primary analysis followed the intention-to-treat principle and included all randomly assigned patients, except those randomly assigned in error or who withdrew consent with explicit refusal of data use. A per-protocol

analysis excluding patients with major protocol deviations was performed as a supportive analysis, and sensitivity analyses were performed to assess the effect of missing outcomes within the same analytical set than the primary intention-to-treat analysis.

Non-inferiority was evaluated hierarchically with a one-sided type I error of 0.025. The primary endpoint was tested first, with a non-inferiority margin of 1.25 for the hazard ratio (HR). Non-inferiority was concluded if the upper bound of the two-sided 95% CI was less than 1.25. Conditional on establishing non-inferiority for the primary endpoint, the key secondary endpoint was tested with a non-inferiority margin of 1.40. No further adjustment for multiplicity was made. The selected

	β -blocker discontinuation (n=3092)	β -blocker continuation (n=3146)
Demographic and cardiovascular risk		
Age, years	63.3 (56.0–70.6)	63.0 (56.0–70.6)
Sex		
Male	2618 (84.7%)	2657 (84.5%)
Female	474 (15.3%)	489 (15.5%)
BMI, kg/m ²	25.8 (23.7–28.4)	25.8 (23.7–28.4)
Current smoker	751 (24.3%)	694 (22.1%)
Hypertension	1540 (49.8%)	1571 (49.9%)
Diabetes	873 (28.2%)	843 (26.8%)
Dyslipidaemia	1733 (56.0%)	1782 (56.6%)
Medical history		
ST-segment elevation myocardial infarction	1882 (60.9%)	1891 (60.1%)
Duration between last myocardial infarction and random assignment, years	3.6 (1.6–7.4)	3.6 (1.5–7.5)
Multivessel disease	1707 (55.2%)	1737 (55.2%)
Revascularisation for index myocardial infarction (PCI or CABG)	2963 (95.8%)	3011 (95.7%)
Peripheral vascular disease	115 (3.7%)	95 (3.0%)
Stroke or transient ischemic attack	86 (2.8%)	104 (3.3%)
Health status		
Ejection fraction at randomisation	59.4% (53.0–62.0)	59.0% (53.0–63.0)
Systolic blood pressure at baseline, mm Hg	130 (120–140)	130 (120–140)
Diastolic blood pressure at baseline, mm Hg	75 (69–81)	75 (69–82)
Heart rate at baseline, bpm	67 (60–75)	67 (60–75)
Data are median (IQR) or n (%). Percentages and summary statistics were calculated among patients with available data. Missing values were 228 of 6238 for BMI, 15 of 6238 for time from index myocardial infarction to random assignment, 80 of 6238 for left ventricular ejection fraction, 91 of 6238 for systolic blood pressure, 91 of 6238 for diastolic blood pressure, and 98 of 6238 for heart rate. CABG=coronary artery bypass grafting. PCI=percutaneous coronary intervention.		

Table 1: Baseline characteristics of the patients

margins were consistent with those used in the original designs of the parent trials and with the non-inferiority boundaries commonly used in cardiovascular trials. Because only two trials contributed to this pooled analysis, a one-stage mixed-effects Cox proportional hazards model with trial included as a random effect was retained as the primary analytical approach. Two-stage analyses were performed as sensitivity analyses and were considered exploratory.

Between-trial heterogeneity was evaluated with the I^2 statistic and with an exploratory test for treatment-by-trial interaction within the one-stage model. Quality-of-life outcomes were analysed with linear mixed-effects models including trial as a random effect and adjustment for baseline values. All statistical analyses were performed, and graphs were generated, with SAS software, version 9.4 (SAS Institute), or R (version 4.3.2) for the two-stage random-effects analyses.

Role of the funding source

There was no funding source for this study.

Results

This pooled analysis of individual patient data comprised data from 6238 patients (3698 from ABYSS and 2540 from SMART-DECISION). In ABYSS, patients with a history of myocardial infarction at least 6 months before randomisation and receiving β -blocker therapy were enrolled from 49 centres in France between Aug 28, 2018, and Sept 12, 2022 and followed-up for a median of 3.0 years (IQR 2.0–4.0). In SMART-DECISION, patients treated with β -blockers for at least 1 year after myocardial infarction were enrolled from 25 centres in South Korea between April 27, 2021, and April 7, 2023 and followed-up for a median of 3.1 years (IQR 2.5–3.5). Overall, 3092 (49.6%) of 6238 patients were randomly assigned to β -blocker discontinuation and 3146 (50.4%) of 6238 patients to β -blocker continuation. 685 (11.1%) of 6158 of patients (LVEF was missing for 80 patients) had a mildly reduced LVEF (40–49%). Baseline characteristics of the pooled population were balanced between randomised groups (table 1; appendix p 8). The median age was 63.2 years (IQR 56.0–70.6), 963 (15.4%) of 6238 patients were female, and 5275 (84.6%) were male. ST-segment elevation myocardial infarction accounted for 3773 (60.5%) of 6238 of index events. Median time from index myocardial infarction to randomisation was 3.6 years (IQR 1.6–7.5). Use of cardiovascular medications at the time of randomisation is summarised in the appendix (p 8); bisoprolol was the most prescribed β -blocker in 3465 (55.6%) of 6236 patients with available type of β -blocker at baseline, followed by carvedilol in 1250 (20.0%) patients, and nebivolol in 723 (11.6%). All β -blockers were administered orally. Median total daily doses were 2.5 mg (IQR 2.5–5.0) for bisoprolol, 8.0 mg (6.25–12.5) for carvedilol, and 2.5 mg (2.5–5.0) for nebivolol (appendix p 10). Dosing schedules were not harmonised across trials and are therefore not reported. Median follow-up for the pooled patient population was 3.0 years (2.3–3.8), corresponding to 9276 patient-years of follow-up in the β -blocker discontinuation group and 9438 patient-years in the β -blocker continuation group. The median resting heart rate at the time of randomisation was 67 beats per minute (60–75). Information on race and ethnicity was not harmonised across studies.

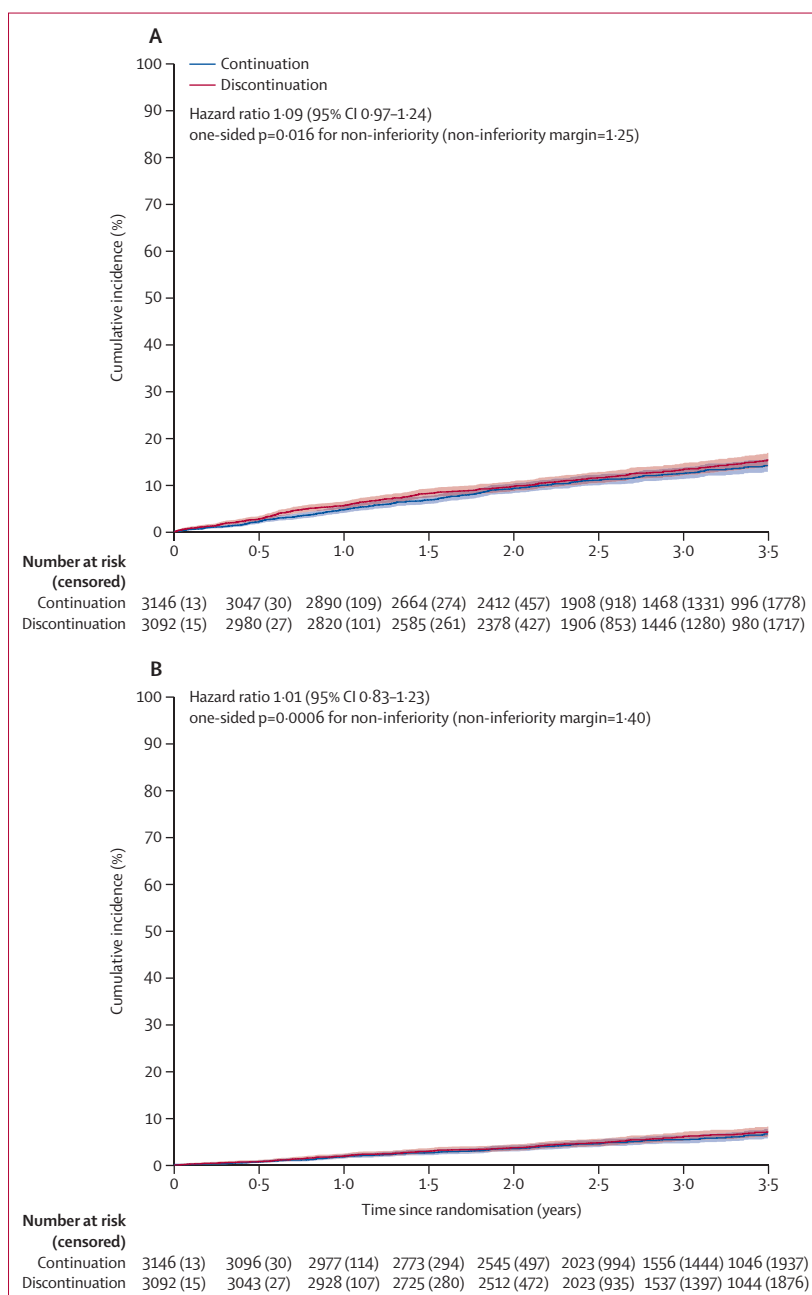


Figure 1: Kaplan-Meier estimates for the cumulative incidence of the primary composite endpoint of death, myocardial infarction, stroke, or hospitalisation for cardiovascular reasons (A) and the key secondary composite endpoint of death, myocardial infarction, or heart failure hospitalisation (B)

	Total (n=6238)	β-blocker discontinuation (n=3092)	β-blocker continuation (n=3146)	Risk difference* (95% CI)	Hazard ratio† (95% CI)	Non-inferiority p value‡
Primary endpoint						
Death, myocardial infarction, stroke, or hospitalisation for cardiovascular reason	1032 (16.5%)	532 (17.2%)	500 (15.9%)	0.50 (−1.19 to 2.20)	1.09 (0.97 to 1.24)	0.016
Key secondary endpoint						
Death, myocardial infarction, or hospitalisation for heart failure	402 (6.4%)	201 (6.5%)	201 (6.4%)	−0.03 (−1.24 to 1.18)	1.01 (0.83 to 1.23)	0.0006
Other secondary endpoints						
Death, myocardial infarction, or stroke	387 (6.2%)	189 (6.1%)	198 (6.3%)	−0.27 (−1.46 to 0.92)	0.96 (0.79 to 1.17)	..
Death, myocardial infarction, stroke, or hospitalisation for heart failure	448 (7.2%)	220 (7.1%)	228 (7.2%)	−0.27 (−1.54 to 1.01)	0.98 (0.81 to 1.17)	..
Death or hospitalisation for heart failure	289 (4.6%)	140 (4.5%)	149 (4.7%)	−0.34 (−1.38 to 0.71)	0.95 (0.75 to 1.20)	..
Death, myocardial infarction, stroke, or coronary revascularisation	569 (9.1%)	282 (9.1%)	287 (9.1%)	−0.33 (−1.73 to 1.06)	0.99 (0.84 to 1.17)	..
Individual components of endpoints						
Death	216 (3.5%)	103 (3.3%)	113 (3.6%)	−0.36 (−1.25 to 0.54)	0.92 (0.70 to 1.20)	..
Myocardial infarction	138 (2.2%)	71 (2.3%)	67 (2.1%)	0.17 (−0.57 to 0.90)	1.07 (0.77 to 1.49)§	..
Stroke	64 (1.0%)	26 (0.8%)	38 (1.2%)	−0.37 (−0.89 to 0.15)	0.69 (0.42 to 1.14)§	..
Hospitalisation for cardiovascular reasons	821 (13.2%)	427 (13.8%)	394 (12.5%)	0.60 (−0.92 to 2.12)	1.11 (0.97 to 1.28)§	..
Hospitalisation for heart failure	93 (1.5%)	50 (1.6%)	43 (1.4%)	0.25 (−0.35 to 0.85)	1.18 (0.78 to 1.77)§	..
Coronary revascularisation (PCI or CABG)	264 (4.2%)	131 (4.2%)	133 (4.2%)	−0.06 (−1.04 to 0.93)	1.00 (0.78 to 1.27)§	..

Data are n (%), unless otherwise specified. Wald-type 95% CIs are presented. Confidence intervals for secondary endpoints have not been adjusted for multiplicity and should therefore be regarded as descriptive. CABG=coronary artery bypass grafting. PCI=percutaneous coronary intervention. *Risk difference in proportion calculated with a generalised linear mixed model and including trial as a random effect. †Hazard ratio estimated with a one-stage mixed-effects Cox proportional-hazards model and included trial as a random effect. ‡Non-inferiority was assessed using a one-sided type 1 error of 0.025, with margins corresponding to an upper 97.5% confidence limit of 1.25 for the primary endpoint and 1.40 for the key secondary endpoint, as specified in the statistical analysis plan. §Cause-specific proportional hazards regression model was used with death as competing risk.

Table 2: Primary and secondary endpoints in the intention-to-treat population

Treatment crossover during follow-up was infrequent, occurring in 271 (4.3%) of 6238 patients; it occurred more often in patients assigned to β-blocker discontinuation (188 [6.1%] of 3092) than in those assigned to continuation (83 [2.6%] of 3146; appendix p 4).

The primary composite endpoint occurred in 532 (17.2%) of 3092 patients assigned to β-blocker discontinuation and in 500 (15.9%) of 3146 patients assigned to β-blocker continuation, corresponding to event rates of approximately 5.7 per 100 patient-years (discontinuation) and 5.3 events per 100 patient-years (continuation). In the pooled one-stage mixed-effects Cox proportional hazards model, including trial as a random effect, β-blocker discontinuation was associated with an HR of 1.09 (95% CI 0.97–1.24), with an absolute risk difference of 0.50 percentage points (−1.19 to 2.20), meeting the criterion for non-inferiority ($p=0.016$; figure 1A; table 2).

The key secondary endpoint occurred in 201 (6.5%) of 3092 patients in the β-blocker discontinuation group and 201 (6.4%) of 3146 patients in the β-blocker continuation group, corresponding to approximately 2.2 events per 100 patient-years (discontinuation) and 2.1 events per 100 patient-years (continuation). The HR for the key secondary endpoint of β-blocker discontinuation versus continuation was 1.01 (95% CI 0.83 to 1.23; absolute risk difference −0.03 percentage points [−1.24 to 1.18]), meeting the criterion for non-inferiority ($p=0.0006$; figure 1B; table 2).

All-cause death occurred in 103 (3.3%) of 3092 patients in the β-blocker discontinuation group and 113 (3.6%) of 3146 patients in the continuation group (HR 0.92 [95% CI 0.70–1.20]). Myocardial infarction occurred in 71 (2.3%) of 3092 patients in the discontinuation group and 67 (2.1%) of 3146 patients in the continuation group; stroke occurred in 26 (0.8%) of 3092 patients in the discontinuation group and 38 (1.2%) of 3146 patients in the continuation group. Hospitalisation for cardiovascular reasons occurred in 427 (13.8%) of 3092 patients assigned to β-blocker discontinuation versus 394 (12.5%) of 3146 patients assigned to β-blocker continuation. Hospitalisations for heart failure occurred in 50 (1.6%) of 3092 patients assigned to β-blocker discontinuation versus 43 (1.4%) of 3146 patients assigned to β-blocker continuation. Trial-specific detailed causes of cardiovascular hospitalisations by treatment group are provided in the appendix (p 11).

During the first 12 months after randomisation, the Kaplan–Meier estimated cumulative incidence of the primary endpoint was 5.6% in the β-blocker discontinuation group and 4.7% in the continuation group (HR 1.19 [95% CI 0.95–1.48]; figure 2). After 12 months, the HR was 1.01 (0.85–1.21). For the key secondary endpoint, event rates during the first 12 months were low and similar between groups, with Kaplan–Meier estimates of 1.9% in the discontinuation group and 1.8% in the continuation group (1.06

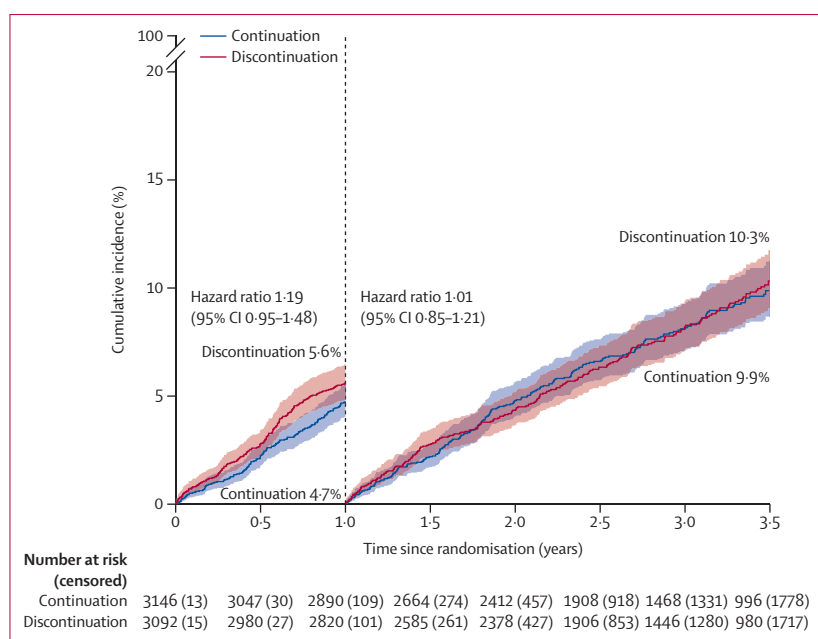


Figure 2: Landmark analysis at 12 months for the primary composite endpoint of death, myocardial infarction, stroke, or hospitalisation for cardiovascular reasons

[0.73–1.53]). After 12 months, the HR was 1.07 (0.83–1.38), with no evidence of a differential treatment effect across time periods (appendix p 5).

No evidence of heterogeneity in treatment effect was observed across exploratory subgroup analyses for the primary and key secondary endpoints, including for sex, type of myocardial infarction, β -blocker type, and LVEF (figure 3; appendix p 6). Of the 6238 patients, 685 (11.1%) had mildly reduced LVEF (40–49%). In this subgroup, the primary endpoint occurred in 63 (19.0%) of 331 patients in the β -blocker discontinuation group and 64 (18.1%) of 354 patients in the β -blocker continuation group (HR 0.97 [95% CI 0.69–1.38]), and the secondary endpoint occurred in 32 (9.7%) of 331 patients in the β -blocker discontinuation group and 36 (10.2%) of 354 in the β -blocker continuation group (0.87 [0.54–1.41]). Among patients with preserved LVEF ($\geq 50\%$), the primary endpoint occurred in 455 (16.7%) of 2721 patients in the β -blocker discontinuation group versus 430 (15.6%) of 2752 patients in the β -blocker continuation group (1.09 [0.96–1.24]), and the secondary endpoint occurred in 166 (6.1%) of 2721 patients in the discontinuation group versus 164 (6.0%) of 2752 patients in the continuation group (1.03 [0.83–1.28]). There was no evidence of heterogeneity across LVEF subgroups ($p_{\text{heterogeneity}}=0.55$ for the primary endpoint and 0.54 for the key secondary endpoint).

When LVEF was modelled as a continuous variable using restricted cubic splines, there was no evidence of interaction between treatment effect and ejection fraction across the spectrum above 40% for the primary endpoint (appendix p 7).

In sensitivity analyses, pooled estimates for both the primary and key secondary endpoints were consistent with the main one-stage analysis. Treatment-by-trial interaction tests did not show statistical evidence of heterogeneity for the primary or key secondary endpoint (p for interaction: 0.082 for the primary endpoint and 0.087 for the key secondary endpoint). In the exploratory two-stage analysis, I^2 was 69% (figure 4). Sensitivity analyses accounting for missing outcome data under missing-at-random and missing-not-at-random assumptions yielded similar findings (appendix p 13). The per-protocol analysis, restricted to 5906 patients without major protocol deviations, was supportive and directionally consistent with the intention-to-treat analysis (appendix p 15). The primary composite endpoint occurred in 464 (16.2%) of 2872 patients assigned to β -blocker discontinuation and 482 (15.9%) of 3034 patients assigned to continuation (HR 1.03 [95% CI 0.90–1.17]; $p_{\text{non-inferiority}}=0.0012$). The key secondary endpoint occurred in 172 (6.0%) of 2872 patients in the discontinuation group and 195 (6.4%) of 3034 patients in the continuation group (0.93 [0.75–1.14]; $p_{\text{non-inferiority}}<0.0001$). Findings were consistent irrespective of β -blocker dosage (appendix p 17).

Baseline EQ-5D-5L scores were similar in the two groups (0.867 [SD 0.162] in the discontinuation group and 0.866 [0.165] in the continuation group). At month 6 and month 12, EQ-5D-5L remained stable with no significant improvement observed with β -blocker discontinuation over continuation during follow-up (appendix p 18).

Discussion

Whether β -blocker discontinuation is necessary in patients with chronic coronary syndrome who survived several years after a myocardial infarction without significant left ventricular dysfunction is a major unresolved clinical question. In this pooled analysis of individual patient data from the ABYSS and SMART-DECISION trials, discontinuation of β -blocker therapy in stable patients after a myocardial infarction met the criteria for non-inferiority for both the primary and the secondary composite cardiovascular endpoints corresponding to the two primary endpoints of the two studies. This analysis provides more robust evidence to support the safety of β -blocker discontinuation in patients who survived several years after a myocardial infarction on β -blocker therapy, and who do not require β -blockers for another indication.

Recent registries and randomised trials conducted in the reperfusion era, along with subsequent pooled analyses, have re-examined the role of β -blockers after myocardial infarction, consistently identifying LVEF as a key determinant of treatment effect.^{1–6,13–16} Benefit was shown in patients with mildly reduced LVEF (40–49%), whereas those with preserved LVEF ($\geq 50\%$) derive

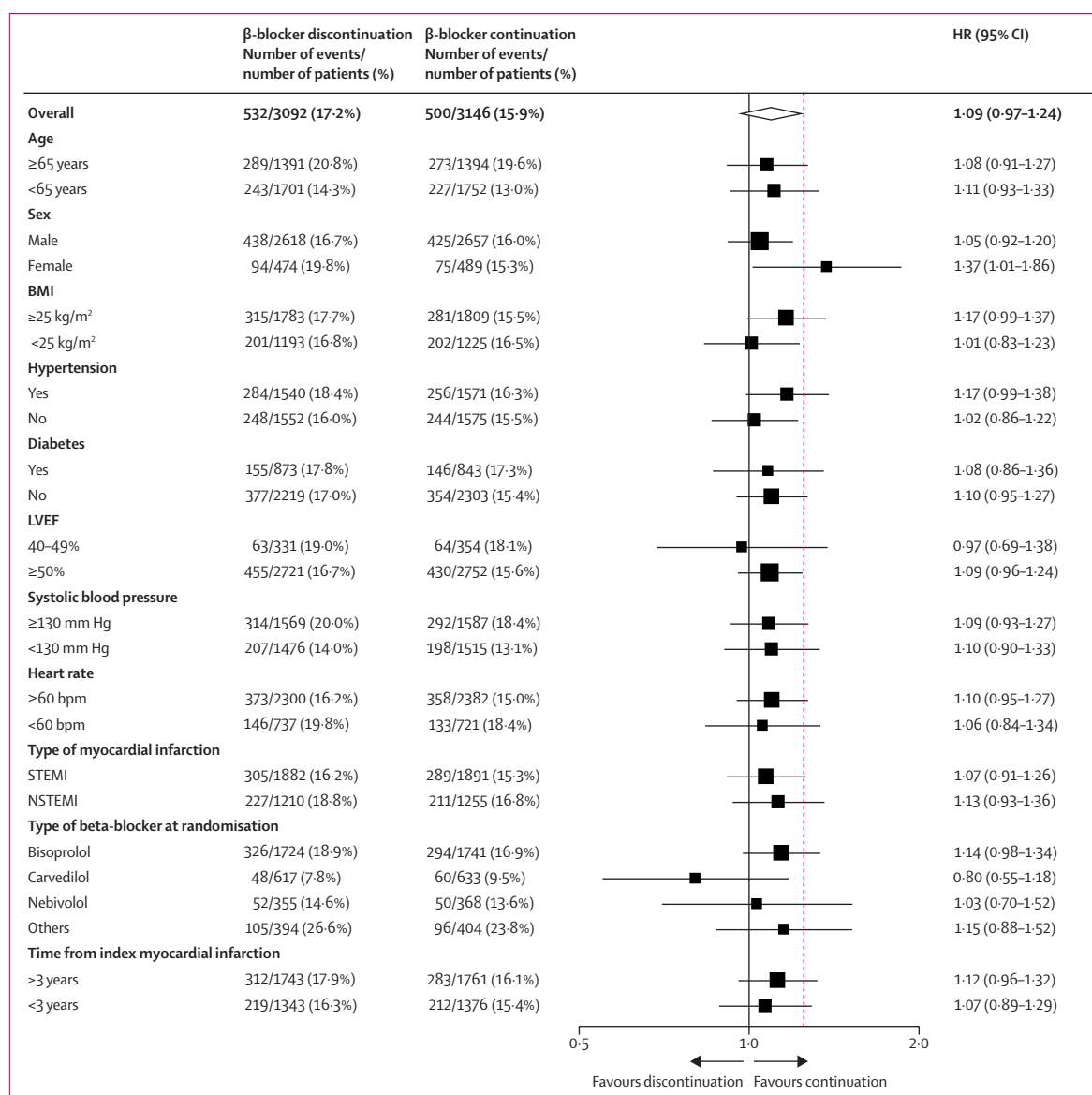


Figure 3: Subgroup analysis for the primary composite endpoint of death, myocardial infarction, stroke, or hospitalisation for cardiovascular reasons

The solid black vertical line indicates an HR of 1 and the dashed red vertical line indicates the non-inferiority margin. HR=hazard ratio. LVEF=left ventricular ejection fraction. NSTEMI=non-ST-segment elevation myocardial infarction. STEMI=ST-segment elevation myocardial infarction.

limited benefit from early initiation. Although exploratory analyses of these trials have suggested the safety of a discontinuation strategy in the early post-myocardial infarction period on long-term outcomes, these studies were not designed to address the distinct question of long-term continuation versus discontinuation in stable patients after myocardial infarction.¹⁷

ABYSS addressed the latter question but could not establish non-inferiority of β -blocker discontinuation for a broad composite endpoint that included cardiovascular hospitalisation. By contrast, SMART-DECISION evaluated a similar population of patients who were stable after a myocardial infarction using a narrower

primary composite of death from any cause, recurrent myocardial infarction, or hospitalisation for heart failure, and found β -blocker discontinuation to be non-inferior to continuation.

Comprising individual patient-level data from 6238 patients in the ABYSS and SMART-DECISION trials, this pooled analysis provides a unified patient-level assessment with greater than either individual trial statistical power showing that a strategy of β -blocker discontinuation met non-inferiority criteria for both the primary and key secondary composite endpoints. Non-inferiority for the primary endpoint was met by a narrow margin, whereas the key secondary endpoint was met

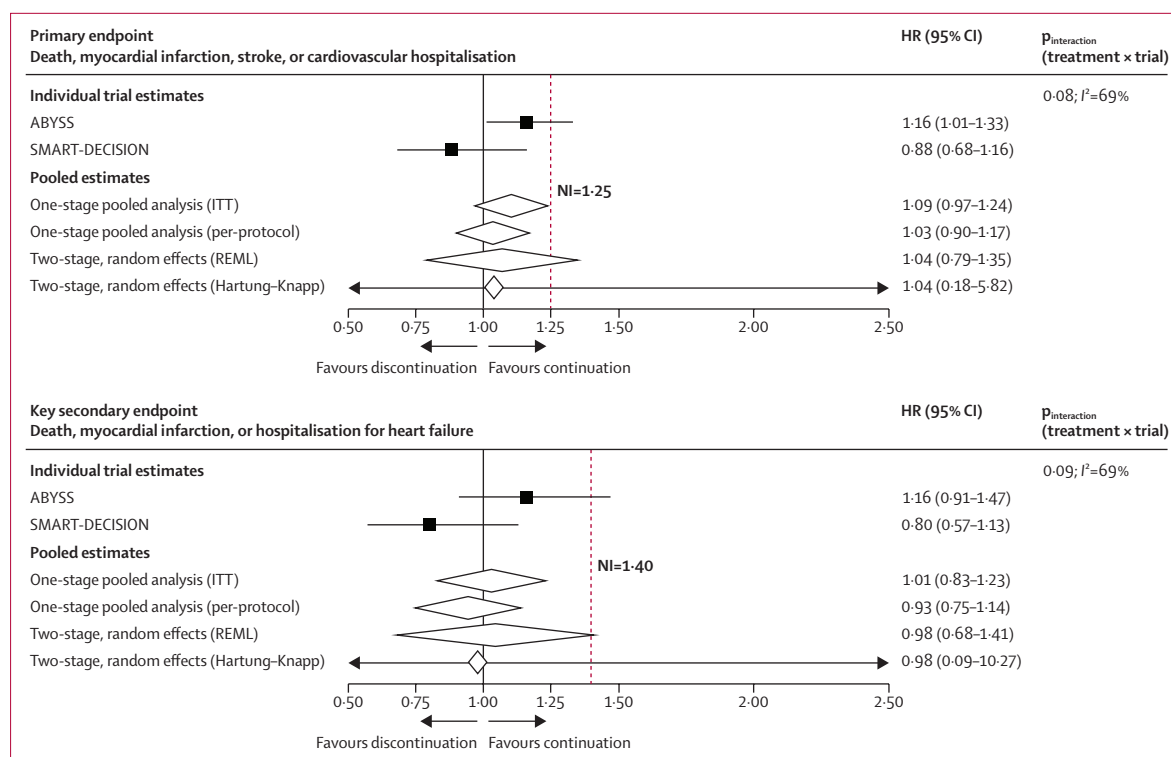


Figure 4: Effect estimates of the primary and key secondary endpoint and its corresponding sensitivity analyses

Data show the composite primary endpoint (death, myocardial infarction, stroke, or cardiovascular hospitalisation) and the key secondary endpoint (death, myocardial infarction, or heart failure hospitalisation). Filled squares represent individual trial estimates (area proportional to weight); diamonds represent pooled estimates. The primary analysis is the one-stage mixed-effects Cox proportional-hazards with trial included as a random effect. Dashed vertical lines indicate non-inferiority margins (primary endpoint: HR 1.25; key secondary endpoint: HR 1.40). For the two-stage random-effects model, the Hartung-Knapp correction (df=1) yields markedly wide confidence intervals; arrowheads indicate bounds exceeding axis limits (primary: 0.18–5.82; key secondary: 0.09–10.27). Interaction p values are from the one-stage model and are exploratory. HR=hazard ratio. ITT=intention to treat. NI=non-inferiority. REML=restricted maximum likelihood.

more robustly, a discrepancy largely driven by differences in cardiovascular hospitalisation, a softer endpoint more susceptible to practice variation and open-label assessment than hard outcomes. Exploratory analyses suggested between-trial heterogeneity, with non-inferiority less apparent in ABYSS than in SMART-DECISION, driven mainly by non-fatal hospitalisations for coronary investigation rather than by an excess of hard outcomes. The higher absolute event rates in ABYSS nonetheless raise the possibility that the safety of β -blocker discontinuation might be less certain in patients at higher risk of adverse cardiovascular outcomes.

Our pooled analysis supports consideration of β -blockers discontinuation in patients with chronic coronary syndrome with previous myocardial infarction, LVEF $\geq 40\%$, no heart failure, and no other indication for β -blocker therapy, with appropriate clinical monitoring. The subgroup of patients with mildly reduced ejection fraction was under-represented in the pooled cohort and findings in this subgroup should be interpreted carefully given the limited power and recent evidence suggesting potential benefit of β -blocker continuation in this population.¹⁰

ABYSS results showed that β -blocker discontinuation resulted in an increase in blood pressure,¹¹ an effect that

was also observed in the SMART-DECISION trial, although to a lesser extent.⁸ Another recent analysis of ABYSS showed an association of higher heart rate with worse cardiovascular outcomes.¹⁸ Such results highlight that β -blockers discontinuation should not be considered as a neutral intervention and there is a need for careful clinical monitoring and appropriate therapeutic adjustment when β -blockers are stopped.

Importantly, neither ABYSS nor SMART-DECISION showed an improvement in quality of life with β -blocker discontinuation, which challenges the assumption that chronic β -blocker therapy is poorly tolerated in this population. This finding was consistent in the present analysis and might relate, in part, to the lower β -blocker doses used in contemporary practice compared with the higher doses tested in earlier (pre-reperfusion era) studies.^{1,2,19} Interestingly, the impact of β -blocker introduction in the acute phase of myocardial infarction that was analysed in the REDUCE-AMI trial also suggested a good tolerance of β -blockers with no effect in the EQ-5D-5L self-reported questionnaire.²⁰

This pooled analysis has limitations. Only two randomised trials were available, reflecting the fact that ABYSS and SMART-DECISION are the only studies

specifically designed to evaluate long-term β -blocker discontinuation long after myocardial infarction. Cardiovascular hospitalisation was retained according to each parent trial's definition without harmonisation and might have been influenced by the open-label design; however, clinical events were adjudicated by independent blinded committees, and results were consistent for harder endpoints. The statistical plan was finalised after SMART-DECISION database lock and preserved original trial hypotheses. The primary endpoint included death from any cause. As non-cardiovascular deaths are not expected to be directly influenced by β -blocker discontinuation, their inclusion could dilute any between-group difference in the cardiovascular components of the endpoint. However, all-cause mortality was similar between groups, with no indication of a differential contribution from non-cardiovascular causes. Although crossover was slightly more frequent in the discontinuation group, the per-protocol analysis provided directional consistent findings with the intention-to-treat analysis. β -blocker type and dosing were not standardised, reflecting real-world practice; however, there was no evidence of interaction between treatment effect and β -blocker dose. Finally, patients included in this analysis were selected according to the eligibility criteria of each trial, which excluded patients with clear ongoing indications for β -blocker therapy, including uncontrolled hypertension in ABYSS and atrial fibrillation in SMART-DECISION. Enrolled patients were mostly male individuals recruited in France and South Korea, limiting the generalisability of findings in female patients and patients from other geographical or ethnic backgrounds, and our findings should not be extrapolated to the broader population of cardiovascular patients.

Contributors

JS, J-YH, KHC, and VM conceived and designed the pooled analysis of individual patient data. AD, DK, and EV reviewed the analytic plan, conducted analyses, and provided tables and figures. All authors contributed to data curation. JS and VM drafted the manuscript with input from AD, DK, GM, J-YH, and KHC. All authors critically revised the manuscript, contributed to data interpretation, and were responsible for the decision to submit the manuscript. EV and DK directly accessed and verified the underlying data reported in the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

JS reports proctoring for Abbott Medical and SMT; personal fees from Biotronik, Sanofi-Aventis, and CSL Behring; non-financial support from Abbott Medical, Biotronik, Medtronic, and Novo Nordisk; and equity in 4P-Pharma. KHC reports funding from the Korean Society of Cardiology, the Korean National Institute of Health, the American Chamber of Commerce, Abbott Vascular, and Amgen. VM reports grants from the ACTION Study Group and the Fédération Française de Cardiologie; and is a member of the TIMI Study Group, which reports grants paid to Brigham and Women's Hospital from Abbott, Abiomed, Amgen, Anthos Therapeutics, ARCA Biopharma, AstraZeneca, Boehringer Ingelheim, Cleerly, Daiichi-Sankyo, Ionis Pharmaceuticals, Janssen Research and Development, MedImmune, Merck, Novartis, Pfizer, Regeneron Pharmaceuticals, Roche, Saghmos Therapeutics, Softcell Medical, The Medicines Company, Verve Therapeutics, and Zora Biosciences. GC reports personal fees from Amgen, AstraZeneca, Abbott, Bayer, Biotronik, Bristol-Myers Squibb,

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Data sharing

De-identified individual patient data from the pooled dataset, along with a data dictionary, will be made available beginning 6 months after publication and for a period of 5 years thereafter. Cardiovascular hospitalisation causes were not harmonised across trials beyond use of each parent trial's adjudicated endpoint definition. Access will be granted to qualified researchers with a bona fide academic interest upon submission of a formal proposal outlining the intended analysis. Approval will be subject to a signed data-access agreement and compliance with the restrictions imposed by the parent trials, including data protection laws. Requests should be directed to both corresponding authors. The study protocol and statistical analysis plan are provided in the appendix. Original parent trial protocols are available at nejm.org.

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